IS 6813 MSBA Capstone Project

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Business Problem

Qx conversion rate
19.7%

Mx conversion rate
12.7%

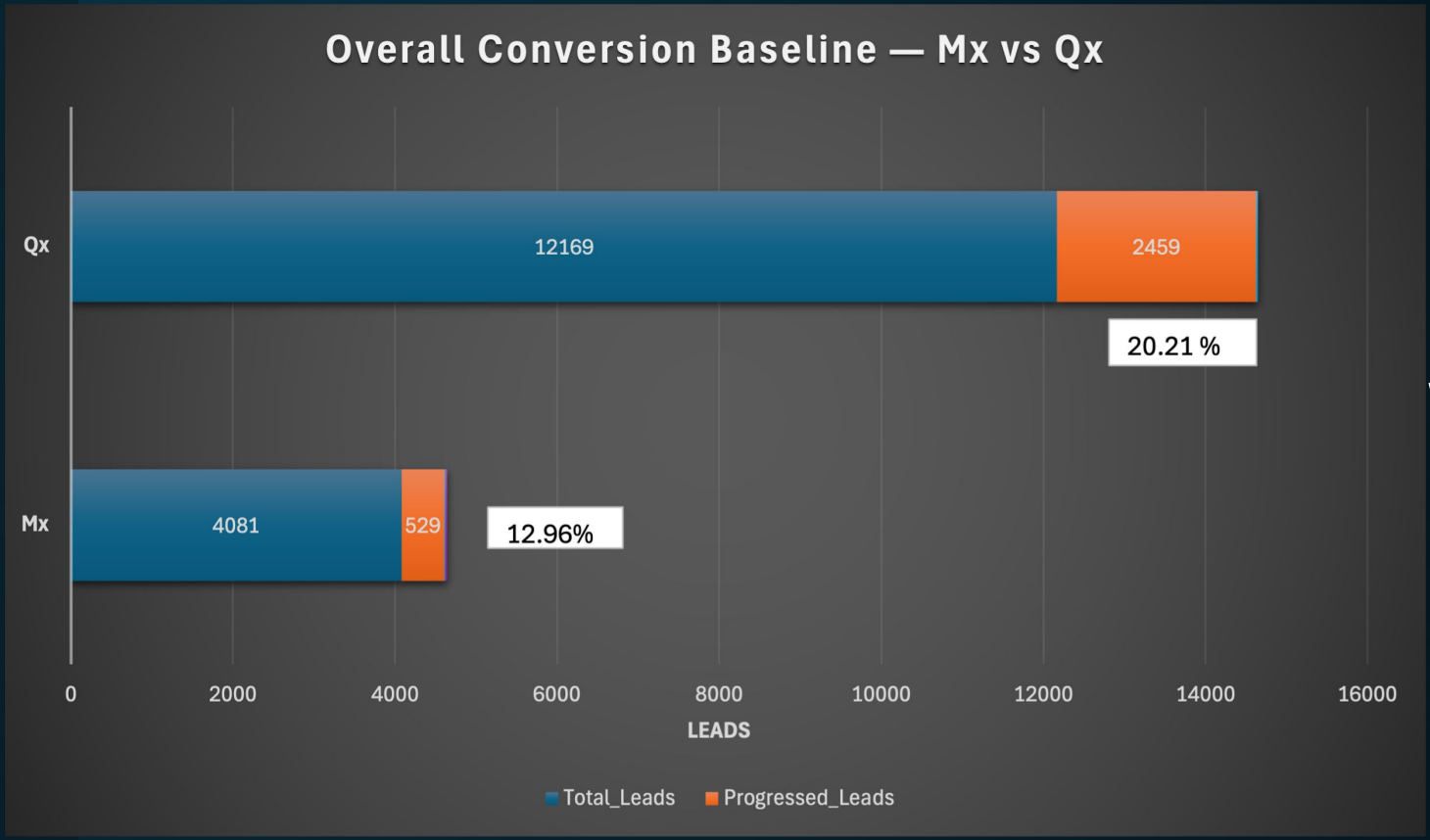
performance gap
(**≈55%** relative difference)

Who is the ideal Mx Customer (ICP)?

✔ **Industries**

✔ **Company Types**

EDA Overview:
Understanding Mx Underperformance



What drives lead progression?

Data Quality & Cleaning

Data Structure

- ✔ 16,644 Rows
- ✔ 14 Variable

Class Imbalance

- ✔ Strong class imbalance with only about 13% of Mx leads progressing
- ✔ The outcome is rare, but valuable.

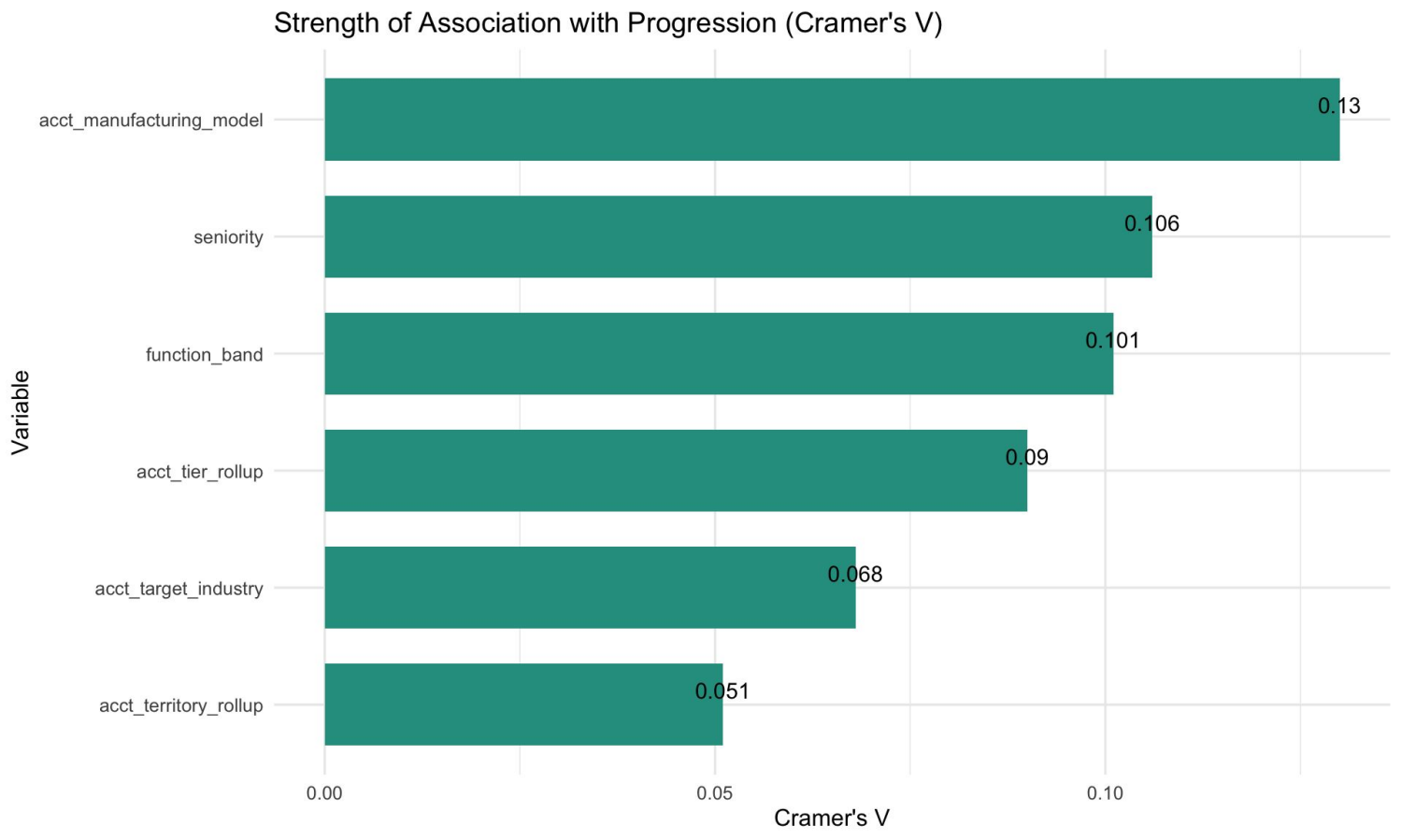
Missing Values

- ✔ Examined potential bias from missing variables and consolidated categorically as “unknown”

Standardized and grouped categorical variables

- ✔ standardized categorical variables (such as grouping job titles into seniority levels and functional area) to make patterns more interpretable.

Relation to Progression



Key Data Insights

Manufacturing Model

- Strongest Predictor of progression
- HOW is more important than WHAT

Buyer Persona

- Directors & Managers drive conversion, not executives

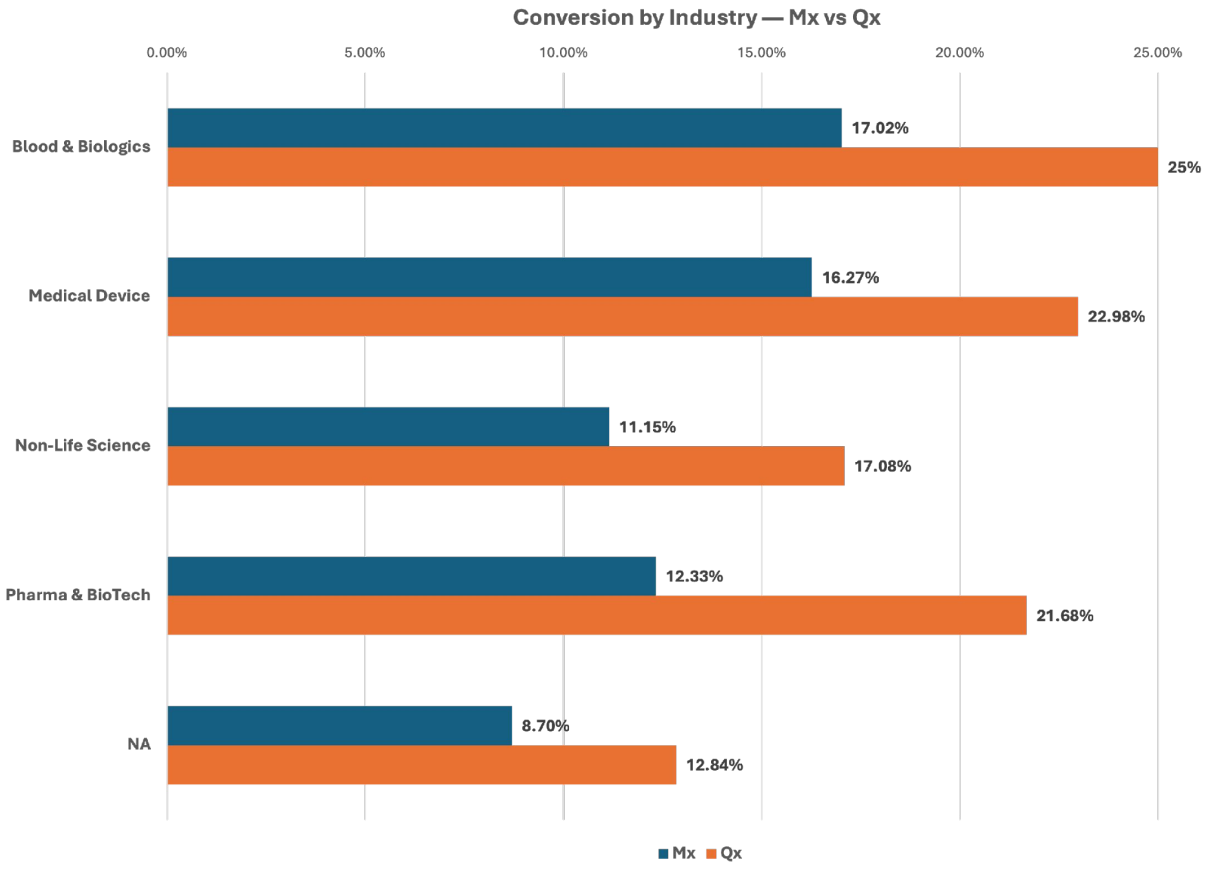
Cross Sale opportunities

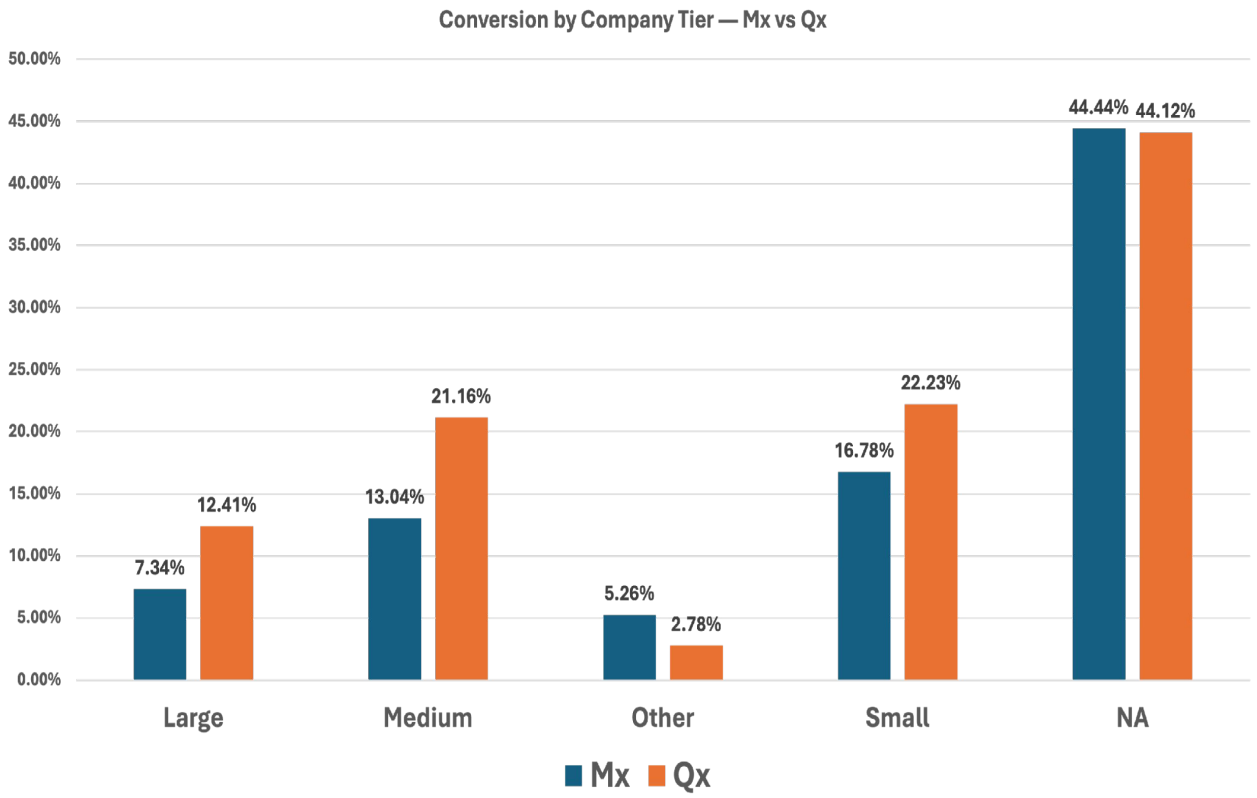
- Some profiles work for QX but not MX
- Cross sale potential for leads that have quality compliance and manufacturing roles, and managers responsible for both

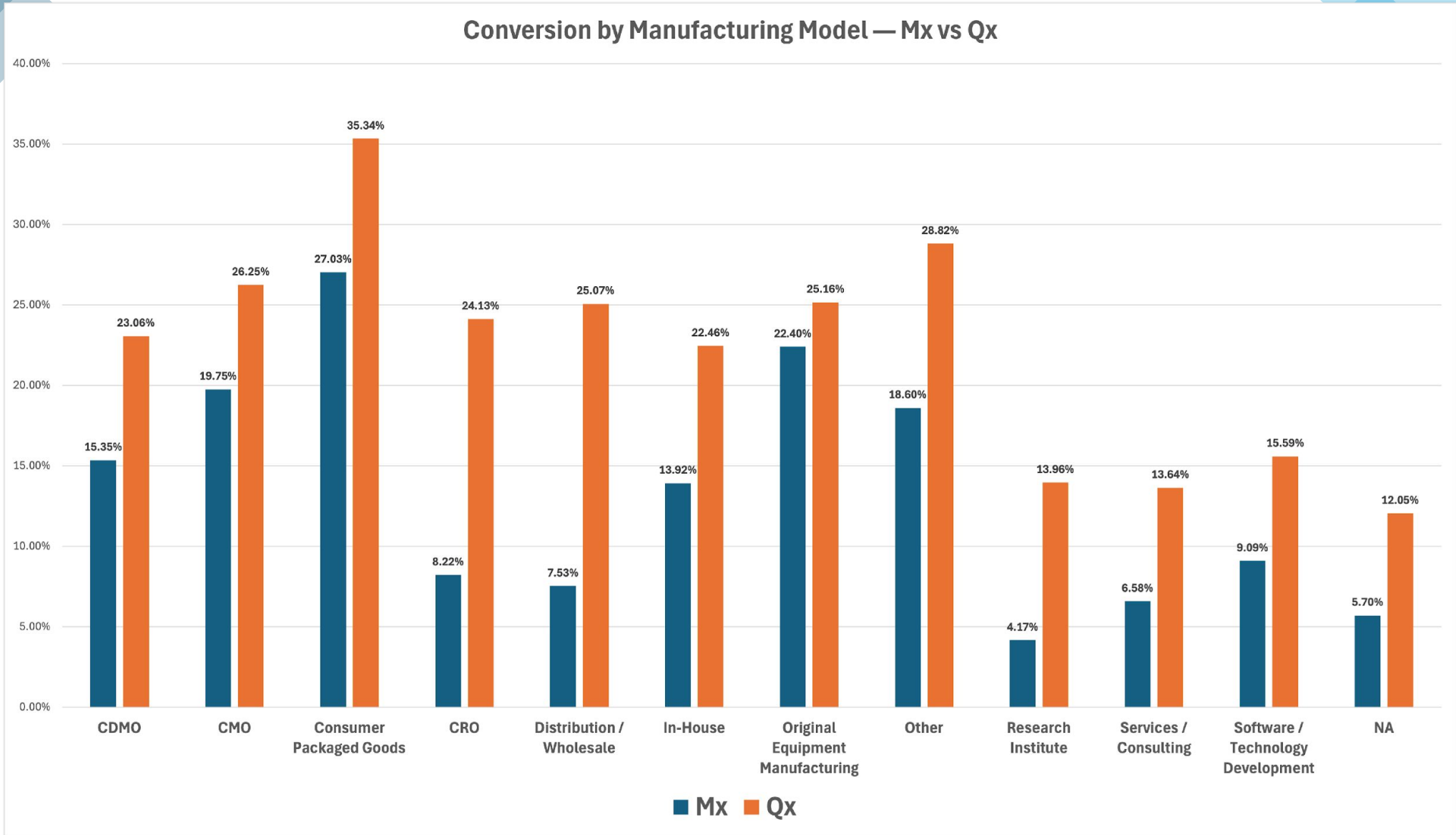
Descriptive Analytics

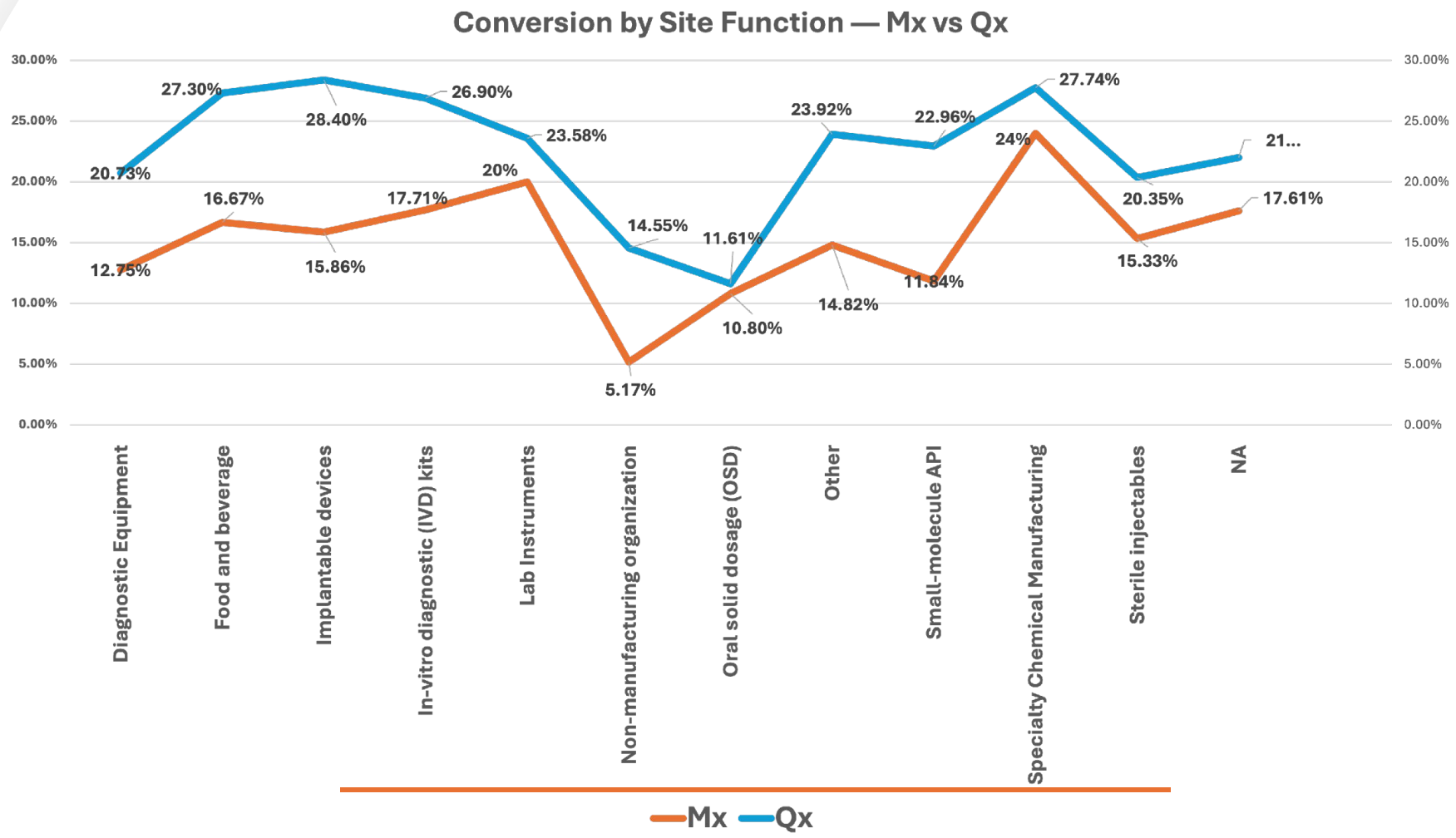
- ✔ **Raw historical data** to understand patterns
- ✔ Identify how **Mx and Qx conversion rates vary**
- ✔ Observe **natural relationships in the data without modeling assumptions**
- ✔ provide the **context for the predictive models introduced later**
- ✔ **which factors may influence progression**

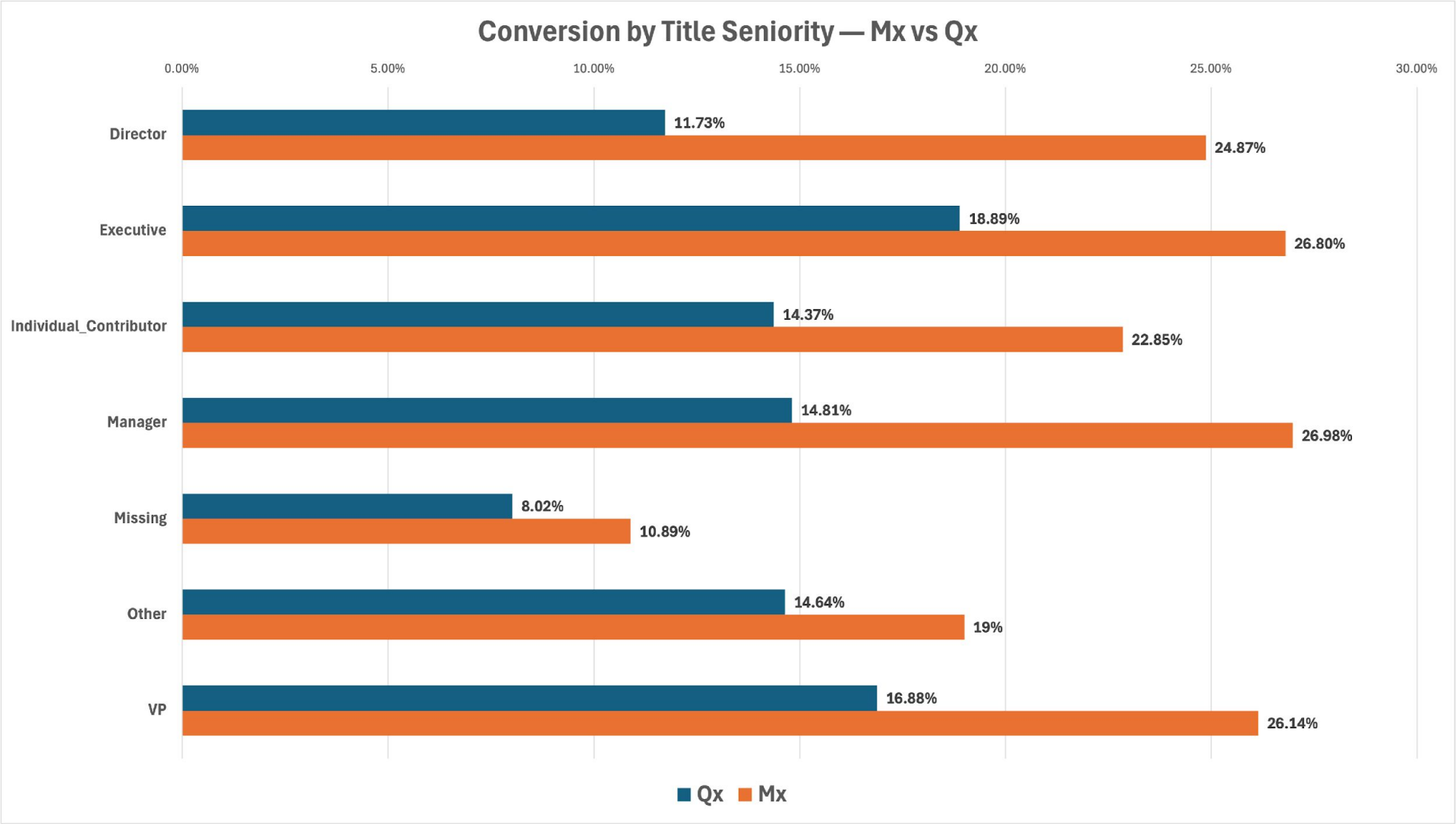
Raw Data Insights

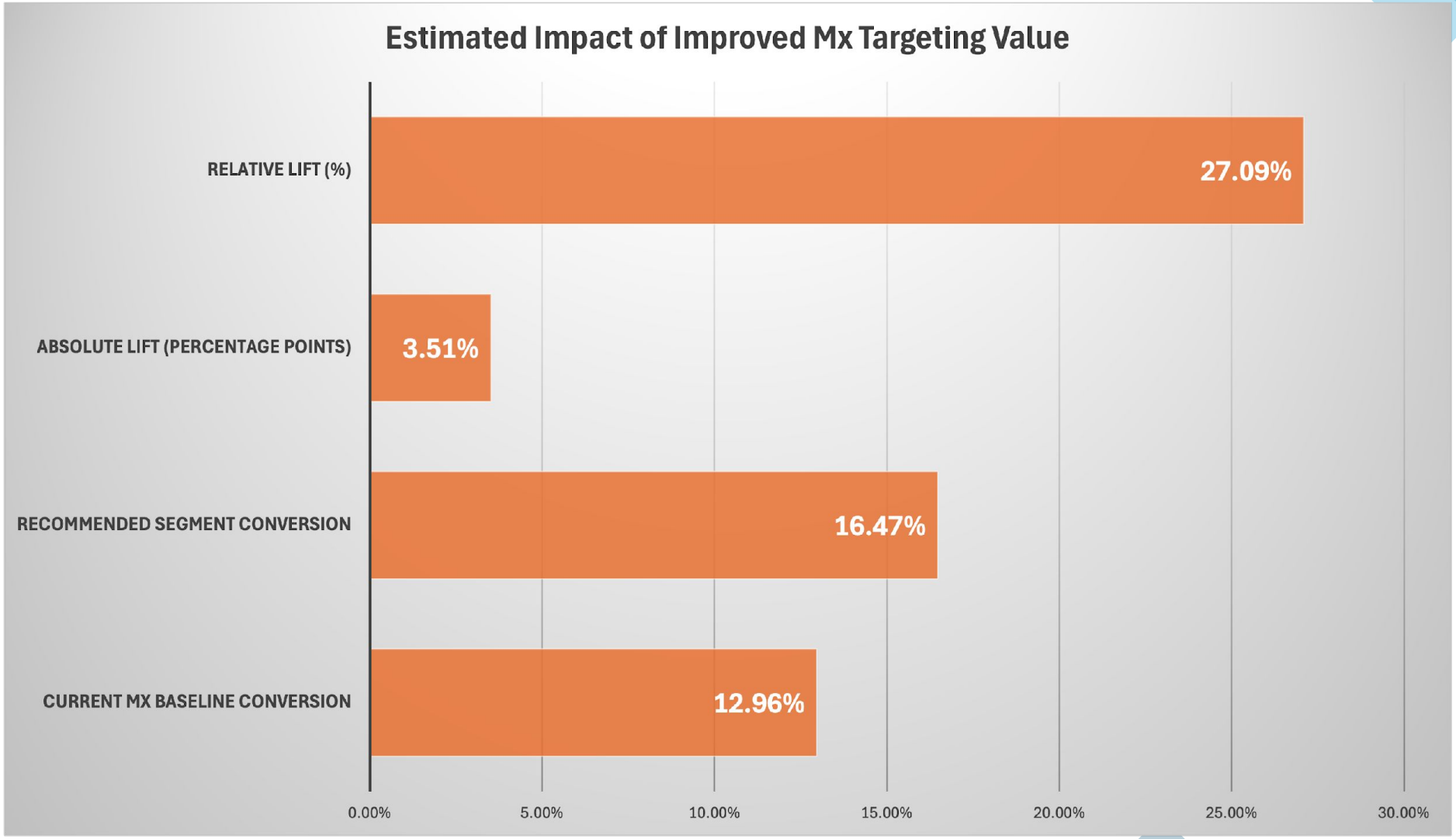





Conversion by Manufacturing Model — Mx vs Qx
40.00%
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
15.35%
23.06%
19.75%
26.25%
27.03%
35.34%
8.22%
24.13%
7.53%
25.07%
13.92%
22.46%
22.40%
25.16%
18.60%
28.82%
4.17%
13.96%
6.58%
13.64%
9.09%
15.59%
5.70%
12.05%
CDMO
CMO
Consumer Packaged Goods
CRO
Distribution / Wholesale
In-House
Original Equipment Manufacturing
Other
Research Institute
Services / Consulting
Software / Technology Development
NA
Mx
Qx

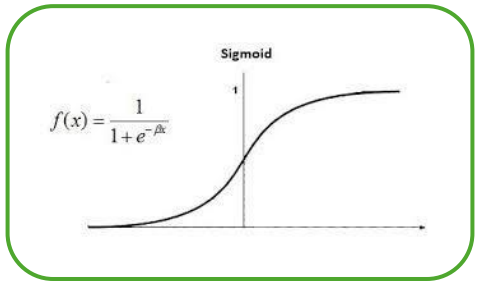

Conversion by Site Function — Mx vs Qx
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
20.73%
27.30%
28.40%
26.90%
23.58%
14.55%
11.61%
23.92%
22.96%
24%
27.74%
20.35%
21...
12.75%
16.67%
15.86%
17.71%
20%
5.17%
10.80%
14.82%
11.84%
15.33%
17.61%
Diagnostic Equipment
Food and beverage
Implantable devices
In-vitro diagnostic (IVD) kits
Lab Instruments
Non-manufacturing organization
Oral solid dosage (OSD)
Other
Small-molecule API
Specialty Chemical Manufacturing
Sterile injectables
NA
Mx
Qx


Conversion by Title Seniority — Mx vs Qx
0.00%
5.00%
10.00%
15.00%
20.00%
25.00%
30.00%
Director
11.73%
24.87%
Executive
18.89%
26.80%
Individual_Contributor
14.37%
22.85%
Manager
14.81%
26.98%
Missing
8.02%
10.89%
Other
14.64%
19%
VP
16.88%
26.14%
Qx
Mx

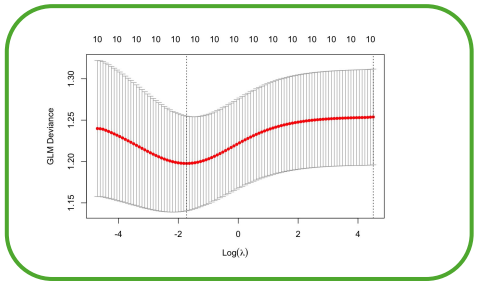

Estimated Impact of Improved Mx Targeting Value
RELATIVE LIFT (%)
27.09%
ABSOLUTE LIFT (PERCENTAGE POINTS)
3.51%
RECOMMENDED SEGMENT CONVERSION
16.47%
CURRENT MX BASELINE CONVERSION
12.96%
0.00%
5.00%
10.00%
15.00%
20.00%
25.00%
30.00%

Predictive Analytics

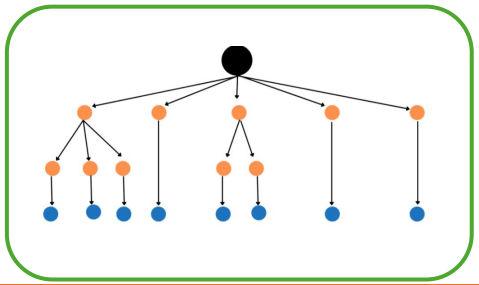
Five predictive models were evaluated:



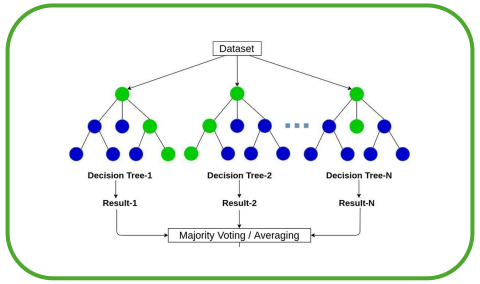
Logistic Regression Model



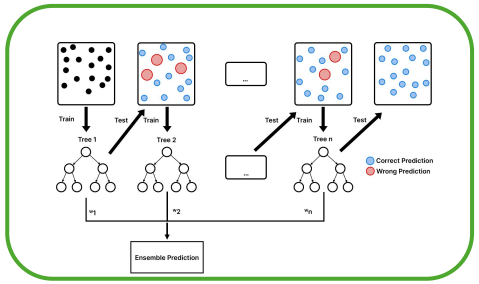
GLMNET logistic regression Model



Decision Tree Model



Random Forest Model



XGBoost Model

Why Logistic Regression ?

Reason	Explanation
High interpretability	Logistic regression clearly shows how each variable (industry, company tier, role, etc.) influences the **probability of progression**.
Directly answers business questions	The model allows us to identify **which industries, account types, and buyer roles are more likely to convert**, aligning with the project objectives.
Transparent for business teams	The outputs are easy to explain to **sales and marketing teams**, making it easier to translate results into targeting strategies.

Understanding Model Outputs

P-value (Statistical Significance)

- Indicates whether a variable has a statistically meaningful relationship with progression.
- Lower p-values suggest stronger evidence that the variable influences conversion.

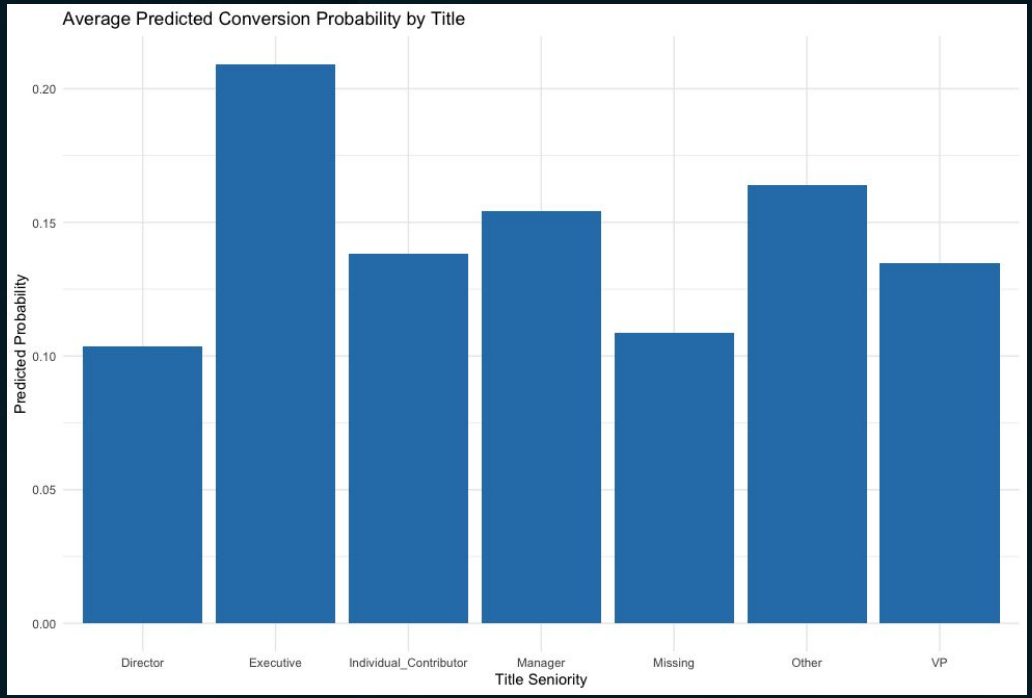
Odds Ratio

Shows **how much a factor increases or decreases the likelihood of progression**.

- Odds Ratio > 1 → increases likelihood of progression
- Odds Ratio < 1 → decreases likelihood of progression

Mx Title Model — Significant Predictors

Predictor	Estimate	Odds Ratio	CI Lower	CI Upper	p-value	Significance
title_seniorityManager	0.4405	1.5534	1.1210	2.1683	0.0088	** Very Significant
title_seniorityExecutive	0.6142	1.8482	1.1078	3.0391	0.0168	* Significant
title_seniorityIndividual_Contributor	0.3394	1.4042	0.9732	2.0292	0.0697	Not Significant
title_functionRegulatory	−1.6978	0.1831	0.0222	1.2180	0.0792	Not Significant
title_seniorityMissing	−0.8364	0.4333	0.1300	1.9750	0.2128	Not Significant



Prescriptive Analytics

- ✔ **Translate predictive insights into actionable strategy** for targeting and outreach.
- ✔ **Recommend where sales and marketing efforts should focus** to improve Mx conversion.
- ✔ Use model insights to **optimize targeting, messaging, and cross-sell opportunities**.
- ✔ Goal: **increase progression rates and maximize conversion impact.**

Mx Sweet Spots

High Priority

- ✔ CMO/CDMO + Small **23.9%**
- ✔ In-House + Small **19.6%**

Low Priority

- ✔ Services **6.5%**
- ✔ Software **<7%**

High-Priority Segments: 32.9%

Current Mx Baseline: 13.5%

Mx Buyer Personas

High Priority		Low Priority	
Executive	18.9%	Engineering/R&D	14.9%
VP	16.9%	Quality/Compliance	15.9%
Manager	14.8%	Software	7.8%

Current Mx Baseline: 13.5% | Best Persona: Executive or Management in Engineering

Cross-Selling

Site Function	Qx Only	Cross Sell	Mx Only
Account size	Small/Medium	Small/Medium	Small/Medium
Title	VP/Head	VP/Head	None
Industry	Medical Devices	Medical Devices	Medical Devices
Manufacturing Model	None	No-Cross sell	CMO

Actionable Recommendations

SDRs

- ✔ Target Small Manufacturers
- ✔ Focus: CMO/CDMO/In-House
- ✔ Avoid Services/Software segments

Sales Leadership

- ✔ Implement lead scoring
- ✔ Track segment performance monthly
- ✔ Validate w/ pilot program

Marketing

- ✔ Shift spend to Contract Manufacturing
- ✔ Reduce Services sector campaigns
- ✔ Focus budgets on Small tier

Expected Business Impact

310 Tier 1 Leads Identified
(9.9% of Mx pipeline)

FROM: 13.5% baseline
TO: 32.9% conversion

+19.4% POINTS
~2.5x improvement potential

Next Step: Pilot → Validate → Scale

Thank you

Questions?